Assimilation of SWOT Altimetry Data for Riverine Flood Reanalysis: From Synthetic to Real Data

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***Abstract*—Floods are one of the most common and devastating natural disasters worldwide. The contribution of remote sensing is important for reducing the impact of flooding both during the event itself and for improving hydrodynamic models by reducing their associated uncertainties. This article presents the innovative capabilities of the Surface Water and Ocean Topography (SWOT) mission, especially its river node products, to enhance the accuracy of riverine flood reanalysis, performed on a 50-km stretch of the Garonne River. The challenge addressed here is quantifying how SWOT river observations, alone and in combination with in-situ gauges, can improve hydraulic parameter estimation and river water level prediction in flood reanalysis. The experiments incorporate various data assimilation strategies, based on the ensemble Kalman filter, which allows for sequential updates of model parameters based on available observations. The experimental results show that while SWOT data alone offers some improvements, combining it with in-situ water level measurements provides the most accurate representation of flood dynamics, both at gauge stations and along the river. The study also investigates the impact of different SWOT revisit frequencies on the model's performance, revealing that assimilating more frequent SWOT observations leads to more reliable flood reanalyses. In the real event, it was demonstrated that the assimilation of SWOT and in-situ data accurately reproduces the water level dynamics, offering promising prospects for future flood monitoring systems. Results show that in the OSSE framework, assimilation reduced water level errors by an order of magnitude, while in the real 2024 event the errors were reduced to below 17 cm, demonstrating the reliability of the approach. This study underscores the complementary role of Earth Observation data in enhancing flood dynamics representation in the riverbed and the floodplains.**

***Index Terms*—Data assimilation, EnKF, Garonne river, riverine floods, SWOT, TELEMAC-2D.**

I. Introduction

FUTURE flood risk is anticipated to rise due to urban growth and unchecked urbanization of floodplains, which collectively contribute to heightened exposure and vulnerability to flooding events [1], [2], [3]. In this context, the integration of satellite-based Earth Observation (EO) data offers substantial potential for improving flood risk forecasting models. Enhanced model performance, supported by EO data, contributes to more effective flood mitigation strategies and the protection of vulnerable infrastructure and assets. In particular, remote sensing (RS) observations of water surface dynamics during flood events provide insights for in-depth hydrodynamic analysis and investigation [4], [5]. Though RS data provide valuable spatially distributed observations, their relatively low revisit frequency limits their utility for resolving the rapid temporal dynamics of flood events. Recent efforts have addressed this limitation through the integration of observations from multiple satellite missions [6], [7], [8], [9], [10]. In contrast, in-situ sensors offer high temporal resolution but are spatially sparse, providing data at only a limited number of locations along river networks. Consequently, neither data source alone is sufficient to fully characterize the spatiotemporal complexity of flood dynamics. Combining them offers a promising path forward, enabling improved representation of flood dynamics through complementary strengths.

In order to fully exploit the complementary strengths of RS and in-situ observations, advanced hydrodynamic models, e.g., with TELEMAC-2D, are required to assimilate these data sources into coherent flood simulations. The TELEMAC-2D [11] code solves depth-averaged free surface flow Navier–Stokes equations, known as Shallow water equations as derived first by Barré de Saint Venant [12]. TELEMAC-2D offers significant advantages in river hydraulics and flood simulation by providing high-fidelity hydrodynamic modeling, while its parallel computing capabilities are suitable for large-scale applications [13], especially in ensemble data assimilation (DA). When being used as a forward model in a DA framework, the model outputs are compared to available observations such as water

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level (WL) measurements and/or flood extent observations [14], [15], [16], [17], [18], [19], [20] to allow for the sequentially updating and correcting parameters and inputs. DA strategies can employ either variational methods or ensemble Kalman filters (EnKF) [21]. EnKF methods, which rely on computing stochastically the forecast error covariance matrices through a limited number of perturbed simulations, can be adapted for the context of free-surface flow [22], [23], [24], resulting in more accurately simulation of real-world flood events.

Similar to in-situ WLs, the assimilation of RS-derived hydrometric WLs is significantly straightforward, especially when compared to the assimilation of flood extents [25]. However, since RS-derived WLs are conventionally obtained by combining the observed flood extents with a digital elevation model (DEM) [26], [27], their retrieval accuracy strongly depends on the quality and resolution of the DEM. Furthermore, because the DEM is used both as an input for the hydraulic model and for deriving observed WLs, an inherent bias arises due to the lack of independence between the observations and model inputs [28]. A review by Grimaldi et al. [29] provides a comprehensive understanding of the value of coarse, medium, and high resolution RS observations for flood extent and WL monitoring. In addition, the review emphasizes the importance of incorporating in-situ data, particularly when hydrodynamic models are utilized. In this regard, altimetry data are highly beneficial for flood reanalysis and forecasting. Water surface elevation (WSE) is measured from space by several nadir satellite altimetry missions, including TOPEX/Poseidon, Jason-1/-2/-3, SARAL/AltiKa, Sentinel-3, and Sentinel-6.

Among satellite altimetry missions, the Surface Water and Ocean Topography (SWOT) wide-swath altimetry satellite represents a significant advancement in oceanography and hydrology [30], even long before its launch in December 2022. It offers global coverage of WSE at high resolution with a 21-day revisit frequency, covering a 120 km-wide swath (with a roughly 20-km gap at nadir). Numerous studies were conducted in preparation for such a mission [31], [32], [33], including the development of data simulators to create algorithms for continental water processing [34]. The merits of SWOT data for the calibration of channel and floodplain roughness, as well as for the estimation of river discharge were demonstrated in [35], [36], [37], [38], [39], [40]. New research opportunities for improving flood forecasting lie in the water depth maps and/or flood extent maps derived from SWOT images. Although the quality of WSE time-series data can be affected by many factors, including land cover, floodplain topography, and channel morphology [36], [41], the extensive applications of satellite altimetry underscore its significant potential across numerous environments.

Many studies have focused on retrieving water depth maps without relying on SWOT or any satellite altimetry data [18], [19], [20], [42], whereas others have conducted preparatory work using simulated SWOT-like observations [31], [32], [33]. More recent efforts have used actual SWOT data to reconstruct discharge and estimate friction parameters [37], [38], [39], [40]. Building on the approach proposed in [43], our contribution demonstrates how SWOT data enhances the representation of both flood extent and river water depth. This study investigates the potential of assimilating SWOT altimetry data for riverine flood applications, progressing from controlled synthetic experiments to the assimilation of real satellite observations. The objective is to assess the merits of SWOT river products to enhance flood reanalysis using the TELEMAC-2D hydrodynamic model. To this end, an observing system simulation experiment (OSSE) is first conducted, employing synthetic EO data generated from a reference hydraulic simulation. This framework serves as a foundation for the subsequent real-data application, allowing for controlled assessment of how SWOT assimilation reduces uncertainties in model inputs and parameters. The OSSE framework also provides the flexibility to test different SWOT revisit frequencies, thereby evaluating their impact on reanalysis skill and highlighting the benefits of more frequent satellite observations for flood forecasting. In the real DA experiment, the focus shifts to identifying methodological adjustments needed to effectively incorporate actual SWOT measurements. Particular attention is dedicated to analyzing the discrepancies between the OSSE and real recent event, which can stem from observational errors, unaccounted for in the synthetical data generation. Although the 2024 event did not result in overbank flooding in the study area, meaningful insights were still gained through analysis of in-channel flow dynamics, demonstrating the relevance of SWOT data even under nonflooding conditions.

The remainder of the article is structured as follows. Section II describes the study area, the hydrodynamic model and observations. Section III presents the method and experimental settings for the OSSE and real DA experiments. Experimental results are presented in Section IV, first in the framework of OSSE (Section IV-B), then for real events (Section IV-C). Finally, conclusions and perspectives are summarized in Section V.

II. Material

A. Study Area and TELEMAC-2D Model

The study area covers a 50-km stretch of the Garonne River in southwest France, extending from Tonneins, downstream of its confluence with the Lot River, to the commune of La Réole (see Fig. 1), historically impacted by floods. This region has a strong agricultural component. Since the 19th century, various infrastructures have been developed to safeguard the Garonne floodplain from major flooding events, such as the historic 1875 flood. Over time, a network of longitudinal dikes and weirs has been gradually built to protect and cultivate floodplains [44].

Fig. 2 depicts the riverbed elevation along the river centerline extracted from the TELEMAC-2D mesh. In order to better analyze results, the river bottom represented by cross sections, acquired from two bathymetric surveys in 2013 and 2023, has also been shown here. These cross sections are provided by the SPC Garonne-Tarn-Lot [44]. They are shown by red and blue crosses in Fig. 2. It reveals that the river bathymetry has changed in some areas where results would differ from simulations.

A hydrodynamic model with a 41 000-node mesh is built with the TELEMAC-2D [11] from bathymetric cross-sectional profiles (see Fig. 2) and topographic data. Boundary conditions (BCs) are provided by a downstream rating curve (La Réole gauge station) and an upstream hydrograph (Tonneins gauge

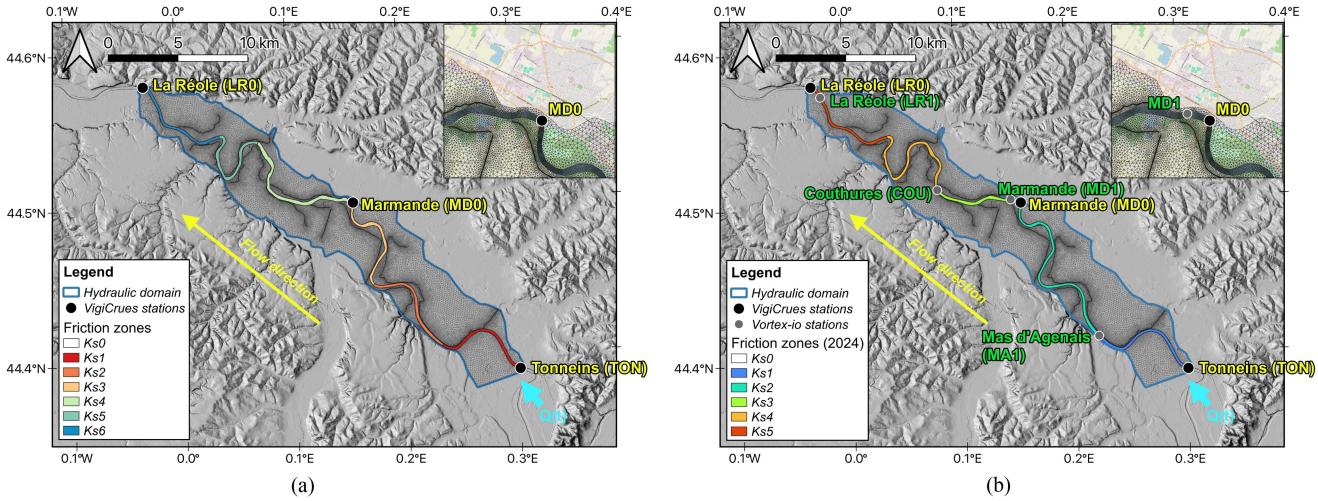


Fig. 1. TELEMAC-2D Garonne Marmandaise domain (within blue outline). The gauge stations are indicated as black (VigiCrues) and gray (Vortex-io) circles. The different river friction zones are indicated as colored segments of the riverbed. The inset figure at the top-right corner magnifies the urban area of Marmande and thereby gauge stations. The left panel is the configuration for OSSE (with six riverbed friction zones) and the right panel is for real event (with five riverbed friction zones). Basemap: © ESRI World Hillshade [45]. (a) OSSE. (b) 2024 real event.

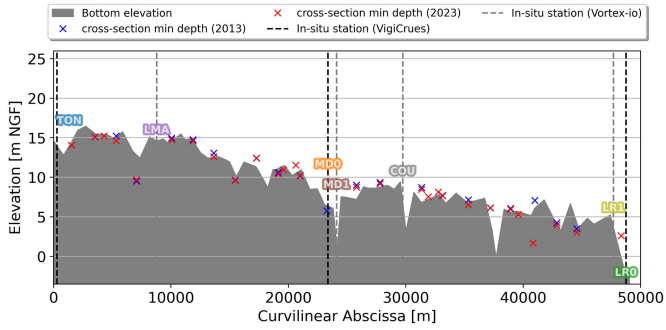


Fig. 2. Bottom elevation profile of the study domain. Blue line is extracted on the centerline from TELEMAC-2D model. Vertical lines represent the location of gauge stations, whereas red and blue crosses represent the bathymetric surveys.

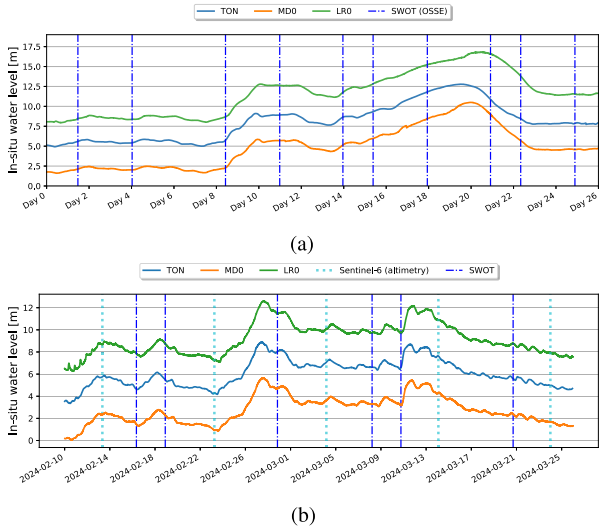


Fig. 3. In-situ WL time-series at VigiCrues stations: Tonneins (blue), Marmande (orange), and La Réole (green) during the (a) synthetical event and (b) 2024 real event. (a) OSSE. (b) 2024 real event.

station). It should be noted that the inflow discharge at Tonneins (Q_{BC}) is one of the control variables in the assimilation process. It is corrected by a multiplying factor μ to the upstream hydrograph. The friction coefficient is defined over several zones, including several segments K_s. A detailed description of the TELEMAC-2D hydrodynamic model is given in [23].

B. In-Situ Data

Observing stations from the national gauge network VigiCrues, are located at Tonneins, Marmande and La Réole, respectively, abbreviated as TON, MD0, and LR0. Fig. 3 depicts the in-situ WLs for the synthetical flood event and the 2024 real event at VigiCrues observing stations, Tonneins (TON), Marmande (MD0) and La Réole (LR0), in blue, orange, and green lines, respectively. Their hydrometric data can be accessed from HydroPortail,[1] a portal of data from the Flood Forecasting Services (SPC) measurement network. The initial calibration of the TELEMAC-2D model was performed using in-situ observations from the three VigiCrues stations, based on a series of nonoverflowing events [46]. Existing works on the same catchment [23], [24] proposed an enhanced setting of the TELEMAC-2D model, where the river channel is divided into six friction zones, using Strickler coefficients [47] denoted by K_{s_i} (with $i \in [1, .., 6]$), as depicted by the solid colored segments in Fig. 1. However, this setup has a limitation related to the number of available in-situ gauge stations, with some zones lacking direct measurements during past flood events, such as K_{s_2} and K_{s_5}.

The SCO-FloodDAM project, initiated in 2021 [48], [49], in collaboration with NASA [50], deployed additional gauge microstations [51] across the study area, which were

[1]https://hydro.eaufrance.fr/

TABLE I
IN-SITU DATA AT VIGICRUES AND VORTEX-IO OBSERVING STATIONS

Stations	Abbre.	Network/ Provider	Installation date	Physical location	Utilization	Time interval	Strickler zone OSSE	Strickler zone Real event
Tonneins	TON	VigiCrues	1989-01-01	Tonneins bridge	Assimilation	15 mins	K_{s_1}	K_{s_1}
Le Mas-d'Agenais	LMA	Vortex-io	2022-11-26	Le Mas-d'Agenais bridge	Assimilation	60 mins	K_{s_2}	K_{s_2}
Marmande	MD0	VigiCrues	1986-03-14	Renaud Jean bridge	Assimilation	15 mins	K_{s_4}	K_{s_3}
Marmande 1	*MD1*	Vortex-io	2022-02-17	D933 road bridge	*Validation*	60 mins	K_{s_4}	K_{s_3}
Couthures-sur-Garonne	COU	Vortex-io	2021-02-18	D3 road bridge	Assimilation	60 mins	K_{s_5}	K_{s_4}
La Réole 1	*LR1*	Vortex-io	2022-04-08	D9 road bridge	*Validation*	60 mins	K_{s_6}	K_{s_5}
La Réole	LR0	VigiCrues	1990-01-01	Rouergue bridge	Assimilation	15 mins	K_{s_6}	K_{s_5}

The stations are listed in order of increasing river kilometer along the study reach.

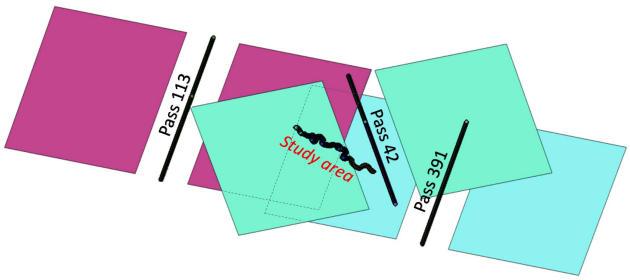


Fig. 4. SWOT passes over the study area.

operational during the 2024 event. Stations at Le Mas-d'Agenais, Marmande, Couthures-sur-Garonne, and La Réole, provided by Vortex-io,[2] contributed to increasing the total number of observing stations to seven, as summarized in Table I in the order of increasing curvilinear abscissa along the study river reach. Benefiting from the improved observation network density and building on the experience from previous studies [23], [24], the number of Strickler zones for the 2024 event was reduced from six to five, as shown in Fig. 1. Consequently, the TELEMAC-2D model configuration for the 2024 event comprises five Strickler zones, each associated with at least one gauge station.

C. RS Data

The SWOT satellite is a pioneering mission designed to provide high-resolution measurements of Earth's surface water. Launched as a collaboration between NASA and CNES, it leverages advanced altimetry technology to map and monitor global surface WLs with unprecedented accuracy. Specifically for inland waters and river monitoring, SWOT can track the height, extent, and dynamics of lakes, rivers, reservoirs, and floodplains [30]. It offers valuable insights into hydrological processes, water management, and climate change impacts by delivering near-real-time information on the world's freshwater resources [36], [37], [40]. The studied 50-km stretch of the Garonne River is covered by three passes 42, 113, and 391, illustrated by Fig. 4. The SWOT acquisition dates during the 2024 event are shown by vertical blue dash-dotted lines in Fig. 3(b).

Sentinel-6 is a Copernicus mission dedicated primarily to monitoring Earth's oceans with exceptional accuracy. Its aims at enhancing the understanding of climate change impacts by providing precise measurements related to sea-level rise, ocean circulation, and climate variability. In addition, Sentinel-6 delivers high-quality observations relevant to continental hydrology. In this study, we analyze data obtained from Sentinel-6's Poseidon-4 Ku/C-band nadir-pointing SAR altimeter. Poseidon-4 operates in high-resolution mode, employing an interleaved chronogram acquisition [52]. Applying the fully focused SAR (FFSAR) processing technique [53] to these measurements enables unprecedented accuracy in nadir altimetry for inland water bodies, achieved through fully coherent processing of radar pulse echoes from the pulse-limited, nadir-looking altimeter. Yet, in order to maintain a robust signal-to-noise-ratio exceeding 20 dB, along-track distance from the satellite track to the river cannot be greater than 2 km. Under these conditions, off-nadir signals enable the estimation of longitudinal profiles of WSEs [54]. As such, this technique demonstrates applicability over an 18-km stretch of the Garonne River within the study area. Sentinel-6 altimetry data were processed using the FFSAR technique [53], providing spatially dense WSE profiles sampled every 10 m along the river centerline at each satellite overpass. Sentinel-6 altimetry data processed using the FFSAR technique have also demonstrated strong potential for monitoring terrestrial water bodies and floods [55], [56], [57], [58], exhibiting reliability even under challenging conditions, such as sandy riverbanks or the presence of hydraulic structures (e.g., locks and dams) [54]. However, the narrow swath coverage of Sentinel-6 remains a limitation, highlighting the necessity to integrate these data with additional EO datasets. Within the study area, Sentinel-6 observations are available only along the aforementioned 18-km section between Le Mas d'Agenais and Marmande. The Sentinel-6 acquisition dates during the 2024 event are indicated by vertical cyan dotted lines in Fig. 3(b).

In this study, SWOT data serve as the primary source, in combination with in-situ data, for DA in the experiments conducted on both the OSSE and the real event. As previously noted, this reach is intersected by SWOT passes 42, 113, and 391. Although passes 42 and 113, only partially cover the hydraulic domain in reality (as shown in Fig. 4), the OSSE setup retains full spatial coverage of the domain for their corresponding SWOT observations, similar to pass 391, in order to maximize the potential benefits of SWOT data for assimilation. Conversely, Sentinel-6 FFSAR measurements are used for validation through longitudinal profiles along the river, while in-situ data provide point observations at the gauge stations.

III. METHOD

Fig. 5 presents the main steps of the proposed methodology. The left side illustrates the steps of the DA algorithm, where

[2]https://www.vortex-io.fr/

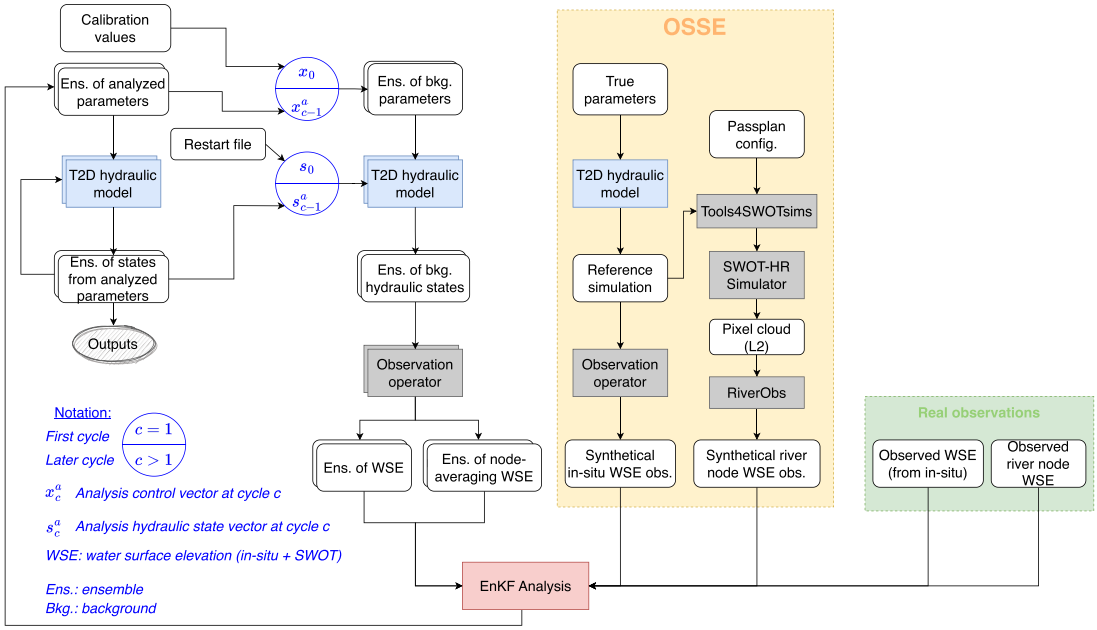


Fig. 5. Workflow of EnKF DA strategy in OSSE (orange box) and real event (green box).

layered boxes represent ensemble-based processes. White boxes denote ensemble inputs, outputs, and variables; gray boxes indicate observation operators; and blue boxes represent T2D model simulations. The orange rectangle highlights the OSSE branch, while the green rectangle corresponds to the real-event (2024) branch using actual observations. Both boxes indicate the starting points of their respective workflows.

A. *Observing System Simulation Experiment*

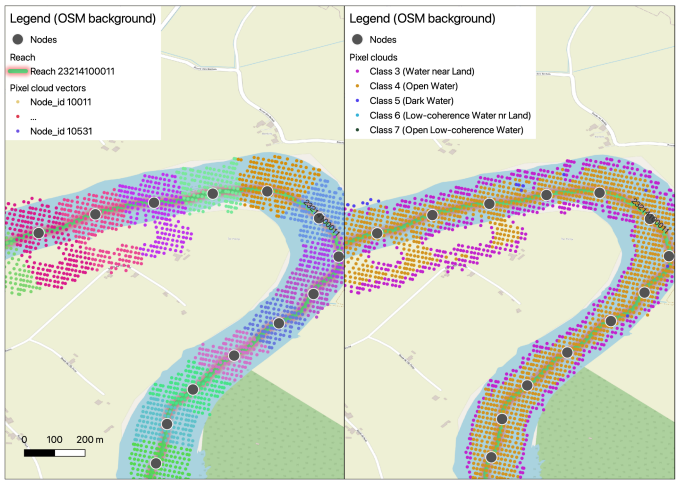
The OSSE observation setting was based on a real flood scenario that occurred in 2021 [22], [23], [24]. Here, the SWOT pass plan (see Fig. 4) was synthetically enhanced, compared to the 21-day repeat cycle of the nominal science orbit, by tripling the number of passes during the flood period [43]. This allows a temporal sampling of 2-3 days for SWOT over the studied catchment, which is more ideal for flood studies.

The synthetical SWOT river products are generated from the reference water depth maps at SWOT overpass times following the configured pass plan. The reference is extracted from re-analysis of the real 2021 flood event [23]. Parameters and water depth in the floodplain have been smoothed to represent a more realistic flood dynamics. The maps are first processed using the Tools4SWOTsims toolbox [34]. This toolbox consists of a set of Python scripts to map 1D/2D hydrodynamic model outputs into 2-D WSE rasters that are compatible with the SWOT-HR simulator,[3] dedicated to hydrological science [59]. The output of the SWOT-HR simulator is SWOT observations in pixel cloud format. Such pixel cloud data are then processed by RiverObs[4] which aggregates the WSEs from a selection of pixels to issue WSEs at nodes every 200 m along river centerline and at river reach every 10 km (which are defined by the SWOT River Database or SWORD [60]). The selection of pixels is based on a class assignment related to the class of the water pixels (i.e., open water, water near land, *dark water*[5], etc.), and weighted by their respective uncertainties to provide node-level WSEs with an error below 10 cm as prescribed in the SWOT requirements. Several factors can contribute to the occurrence of *dark water* in SWOT observations, including calm water surfaces at low incidence angles, signal attenuation or dropout due to rainfall, persistent, or seasonal vegetation cover, and reduced signal-to-noise ratio within the swath, which can lead to undetected water surfaces [30]. The details for SWOT WSE, width, and slope calculation are given in SWOT River Single Pass Product Description Document [61] and example data products.[6]

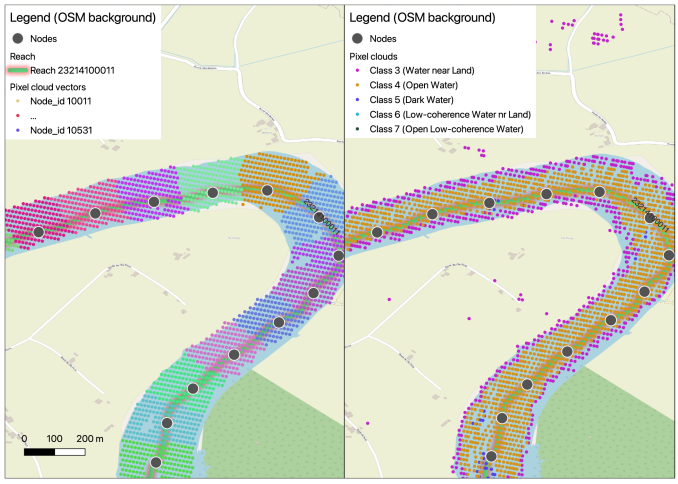
Fig. 6 shows a typical SWOT node-level river product over a meander of a reach of the Garonne River observed on day 8 [third vertical line in Fig. 3(a)] of the OSSE [see Fig. 6(a)] for the synthetical SWOT and on 2024-02-28 [see Fig. 6(b)] for the real one, where each node is an aggregation of pixels. The pixels are color-coded either by their associated node on the left panel, or by their class on the right panel. For each member of the ensemble, the model equivalent of the SWOT node-level WSE is computed using the SWOT-specific observation operator. It is the average of the WSE of certain associated pixels selected according to their quality flag; each pixel WSE is interpolated from the TELEMAC-2D mesh of simulated water depths.

[3]https://github.com/CNES/swot-hydrology-toolbox
[4]https://github.com/SWOTAlgorithms/RiverObs

[5]*Dark water* refers to smooth water surfaces that produce low measurement signals due to specular reflection, wherein the satellite's emitted signal is reflected away from the sensor rather than back toward it.
[6]https://podaac.jpl.nasa.gov/swot?tab=datasets



(a)



(b)

Fig. 6. Node-level WSE SWOT river products: (a) synthetical SWOT data on day 8 (OSSE) and (b) real SWOT data on 2024-02-28 (real event). Left panels: color-coded aggregated pixels for each node; right panels: classification of pixels. Basemap: OpenStreetMap (obtained through QuickMapServices QGIS plugin). (a) OSSE. (b) 2024 real event.

B. DA for OSSE

As for flow rate Q, the DA corrects and updates the friction coefficients K_s in order to approximate as closely as possible the measurements from in-situ and RS sources. Fig. 5 illustrates the assimilation workflow, with key OSSE-specific processes highlighted in the orange box. The reference simulation, which serves as the basis for generating synthetic observations, is configured using parameters derived from prior assimilation of real in-situ data. Dedicated observation operators are employed to derive model equivalents of the observations for each ensemble member. True WSE values are extracted from the reference simulation at corresponding observation times and locations, and are used to generate synthetic in-situ data at the stream-gauge stations.

The deterministic reference simulation driven by the true control vector is referred to as the *Truth*, which is composed of a set of time-varying parameters for friction and inflow. The true value

TABLE II
SUMMARY OF THE OL AND DA EXPERIMENT SETTINGS

Exp. name	Assimilated observations: In-situ WSE	Assimilated observations: SWOT WSE	Control vector (OSSE)	Control vector (real event)
OL	☐	☐	-	-
SWDA	☐	☑	$K_{s_{[0:6]}}, \mu$	$K_{s_{[1:5]}}, \mu$
IDA	☑	☐	$K_{s_{[0:6]}}, \mu$	$K_{s_{[1:5]}}, \mu$
FDA	☑	☑	$K_{s_{[0:6]}}, \mu$	$K_{s_{[1:5]}}, \mu$

for K_{s_k} with $k \in [0, 6]$, μ and δH_k with $k \in [1, 5]$ are obtained from the results of a previous DA experiment on the real 2021 flood event where in-situ observations were assimilated [23]. On the other hand, the deterministic simulation without assimilation, using previously calibrated parameters (that are constant over time), is denoted as the *open loop* (OL). Three EnKF-based experiments are carried out, as detailed in Table II. The IDA experiment assimilates only in-situ WL measurements, SWDA relies solely on synthetic SWOT data, and FDA combines both observation sources. To assess assimilation performance, flood extent maps from four selected time steps—capturing the rising/falling limb and peak of the flood—are extracted from the *Truth* and used as spatial reference benchmarks. The time steps are represented by vertical black-dashed lines in Fig. 3(a).

Within all three DA experiments, the control vector consists of the friction coefficients in the floodplain and riverbed (K_{s_i}, with $i \in [0, 6]$), along with the inflow multiplicative coefficient (μ). For in-situ observations, the proportionality factor τ used to define the observation error is set to 15% , resulting in a standard deviation σ_{obs} equal to 15% of the corresponding observed value.

Much shorter revisit times (i.e. 6, 12, 18, and 24 h) of SWOT data were also evaluated, following SWDA configuration. While these daily and subdaily revisit times do not reflect the actual passage of the SWOT satellite, they are considered for two reasons. First, it is hypothesized that increasing the measurement frequency could help reduce modeling errors. Second, such an approach aims to assess the capacity of the assimilation strategy to integrate new space missions that offer higher temporal resolution in the future, e.g., Sentinel-3 Next Generation Topography (S3NG-TOPO) mission [62].

C. DA for Real 2024 Event

As explained previously in Section II-A, the number of friction zones in the riverbed is reduced to five, whereas the total number of gauge stations is now seven with many Vortex-io micro-stations being available for the 2024 event (see Table I). The study period is 45 days between 2024-02-10 and 2024-03-26. Fig. 3(b) depicts the observed in-situ WL for the event at VigiCrues observing stations, namely Tonneins (TON), Marmande (MD0), and La Réole (LR0), in blue, orange, and green lines, respectively. It should be noted that other in-situ data at Vortex-io observing stations are also available that are not shown here. The overpass times SWOT and Sentinel-6 are indicated as vertical blue dash-dotted lines and cyan dotted lines, respectively. Two hydrograph peaks have been observed with

TABLE III
RMSE [M] OF SIMULATED WLS COMPUTED WITH RESPECT TO *TRUTH* IN-SITU WLS AT OBSERVING STATIONS AND CSI [%] OF SIMULATED FLOOD EXTENT MAPS WITH RESPECT TO *TRUTH* FLOOD EXTENT MAPS

	RMSE against in-situ *Truth* WLs			CSI against *Truth* flood maps			
Exp.	TON	MD0	LR0	Day 17 (rising)	Day 19 (rising)	Day 20 (peak)	Day 23 (recess)
OL	0.260	0.398	0.578	87.95%	81.58%	80.64%	77.52%
SWDA	0.325	0.291	0.229	91.69%	82.15%	73.52%	**80.27%**
IDA	0.052	**0.042**	0.053	92.89%	81.79%	**90.77%**	76.57%
FDA	**0.047**	0.056	**0.051**	**93.07%**	**82.44%**	88.32%	78.92%
SWDA 6h	0.450	0.083	**0.104**	92.39%	81.12%	**95.25%**	76.37%
SWDA 12h	0.428	**0.082**	0.115	92.21%	81.10%	93.47%	76.52%
SWDA 18h	**0.350**	0.087	0.132	92.43%	**81.92%**	91.46%	76.52%
SWDA 24h	0.368	0.110	0.191	**92.50%**	81.51%	90.30%	**78.34%**

The lowest RMSE and highest CSI among the experiments for each column is boldfaced.

more than 2000 $\mathrm{m}^3.\mathrm{s}^{-1}$ on 2024-02-27 and 2024-03-11, but they did not cause overflows into the floodplain. The control vector is now composed of five Strickler coefficients in the main channel (i.e., K_{s_i} with $i \in [1, 5]$) and a corrective multiplicative parameter μ for the discharge. Since the river did not cause overflows, the Strickler coefficient K_{s_0} in the floodplain is not involved in the DA experiments for the real event.

IV. RESULTS AND DISCUSSIONS

This section shows the results of the DA experiments in the control space and in the observation space, in the OSSE (Section IV-B) and the real event Section IV-C). Throughout the section, the observations (or the *Truth* in the case of OSSE) are plotted in black-dashed lines, OL is plotted in orange, IDA is in green, SWDA in red, and FDA in violet for the sake of consistency.

A. Assessment Metrics

The accuracy of the simulated WLs (H^m) is evaluated against in-situ observations (H^o) using the root-mean-square error (RMSE), calculated over the assimilation windows across the entire flood event. The RMSE is computed from the time series sampled at observation times, according to

$$\mathrm{RMSE} = \sqrt{\frac{1}{n_{obs}} \sum_{i=1}^{n_{obs}} (H_i^m - H_i^o)^2}. \quad (1)$$

The RMSE is also computed against Sentinel-6 WSE profiles along the river centerline for each experiment. For the 2-D assessments, the critical success index (CSI) is employed as the 2-D performance metric, calculated over the model domain relative to the *Truth* flood extent maps. The CSI ranges from 0% , indicating no spatial agreement, to 100% , representing a perfect match between simulated and *Truth* flood extents.

B. Results for OSSE Experiments

Fig. 7 illustrates the evolution of the control parameters estimated across the suite of DA experiments conducted in this study, starting with Fig. 7(a) for the original pass plan. Specifically, the friction coefficients K_{s_i} (with $i \in [0, 6]$) for the riverbed and floodplain, as well as the scalar inflow correction μ, are presented. The true parameter values, from the reference simulation, are plotted in black and the calibrated coefficients are indicated by horizontal orange-dashed lines. The bottom panels of Fig. 7 present the temporal evolution of the upstream BC discharge at the TON station, as modified by the analysis inflow correction factor μ. This highlights the degree to which the DA framework adjusts the discharge time series to reconcile model dynamics with observations. The timing of the synthetic SWOT satellite overpasses during the flood event is indicated by vertical dashed lines. These mark the assimilation time steps at which observational updates were introduced, particularly for SWDA, enabling assessment of their impact on parameter estimation and flow dynamics.

Table III summarizes the 1-D quantitative assessment across all OSSE experiments. The RMSEs were computed between simulated and reference WLs at the three VigiCrues observation stations, TON, MD0 and LR0, over the whole duration of the synthetical flood event. In addition, Table III also reports the CSI values computed between the simulated and reference flood extent maps at four selected time steps during the event. The first two time steps (days 17 and 19) correspond to the rising limb of the flood, the third (day 20) captures the flood peak, and the final time step (day 23) is during the falling limb.

1) For original SWOT pass plan: The assessment, in the control space, of the assimilation skill among the three DA experiments revealed notable differences in performance. The IDA experiment, which assimilates only in-situ WL observations, demonstrated high performance, recovering both the true parameter values [see Fig. 7(a)] and the associated hydraulic states (see Table III). The SWDA, relying exclusively on SWOT observations, was unable to retrieve the true parameter values. This emphasizes the limitation of SWDA to fully constrain the system, mainly due to the temporal resolution of the SWOT data. In contrast, the FDA experiment, which assimilated both SWOT and in-situ data, demonstrated a robustness to recover the true system states and parameters, benefiting from complementary information without substantial evidence of observational conflict. Overall, FDA proved to be the most effective approach among the three experiments.

In the context of the OSSE, both IDA and FDA demonstrated the ability to retrieve the true riverbed friction coefficients, albeit with varying degrees of accuracy. This was particularly evident in the downstream segments, especially the fourth and fifth river segments, corresponding to K_{s_4} and K_{s_5}. However, here it can be remarked that the FDA analyses (violet lines) are more closely aligned with the known reference values (black lines) than the IDA analyses (green lines). The added value of SWOT data

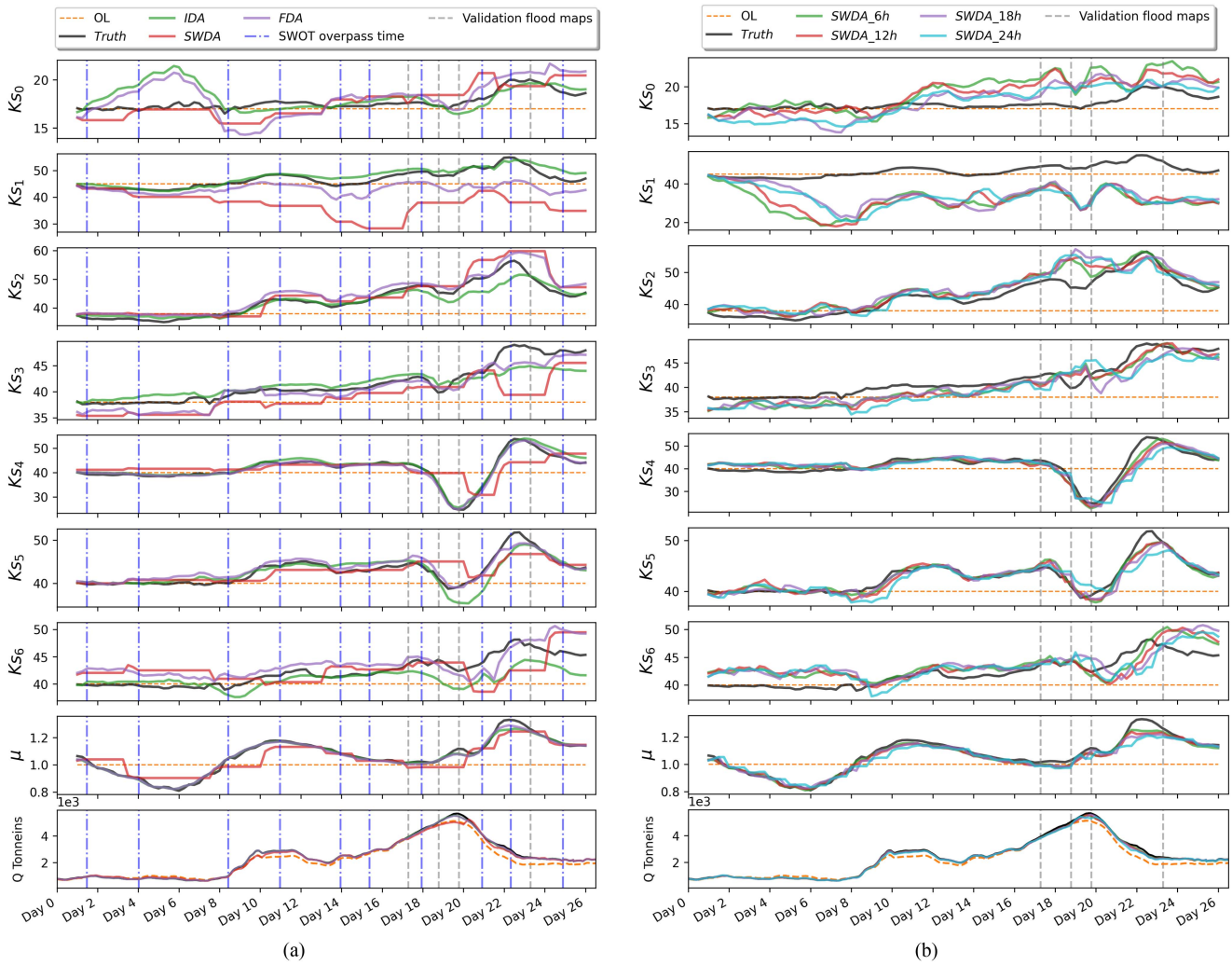


Fig. 7. Analyzed values of the control vector by IDA (green), SWDA (red), and FDA (violet) in (a) OSSE with the original pass plan, and by SWDA at different SWOT revisit times (b) OSSE with the shorter pass plans. From top to bottom: floodplain friction coefficient K_{s_0}, riverbed friction coefficients K_{s_i} for $i \in [1, 6]$, inflow correction factor μ, and upstream forcing $Q(t)$ and the simulated discharge time-series at Tonneins.

in FDA can also be remarked over the analysis of μ during the falling limb, at day 23. Given that the *Truth* was derived from prior model reanalysis (Section III-A), the agreement of both IDA and FDA analyses with the *Truth* can be anticipated. Nevertheless, the SWDA results (red lines) reveal the presence of equifinality in parameter estimation: comparable WL fits at observation stations can be achieved through compensating errors, such as a combination of lower K_{s_3} with higher K_{s_4} values, or vice versa. This underscores the inherent challenges in achieving unique parameter identification using sparse observational data.

The presence of equifinality is also evident in the early-stage (before day 8) analysis of floodplain friction K_{s_0} in IDA and FDA. As water begins to spread across the floodplain, this equifinality diminishes, allowing the floodplain friction estimates to progressively align with the true value, particularly by IDA. Despite these compensatory effects, all analyzed friction coefficients, both within the channel and across the floodplain, remain within physical ranges and are consistently closer to the true parameter values than to the default value, especially near peak flow.

Moreover, the assimilation of in-situ WL observations at Marmande (MD0), located in the fourth river segment, enables highly precise retrieval of K_{s_4} in the IDA and FDA. Here, additional information from SWOT data, in FDA, only slightly improve the parameter estimation during the flood recess (after day 22), indicating that local in-situ data alone can effectively constrain the friction. On the other hand, the FDA analyses for K_{s_3} and K_{s_5} outperform those from IDA, highlighting the advantage of SWOT observations over these ungauged river segments.

Fig. 8 presents the simulated WLs at the TON [see Fig. 8(a)], MD0 [see Fig. 8(b)], and LR0 [see Fig. 8(c)] stations for the OL and all three DA experiments, alongside the *Truth* WLs indicated by black dashed lines. The OL simulation exhibits a slight underestimation of the flood peak across the domain. In contrast, the DA strategies demonstrate a marked improvement in correcting flow propagation within the river channel.

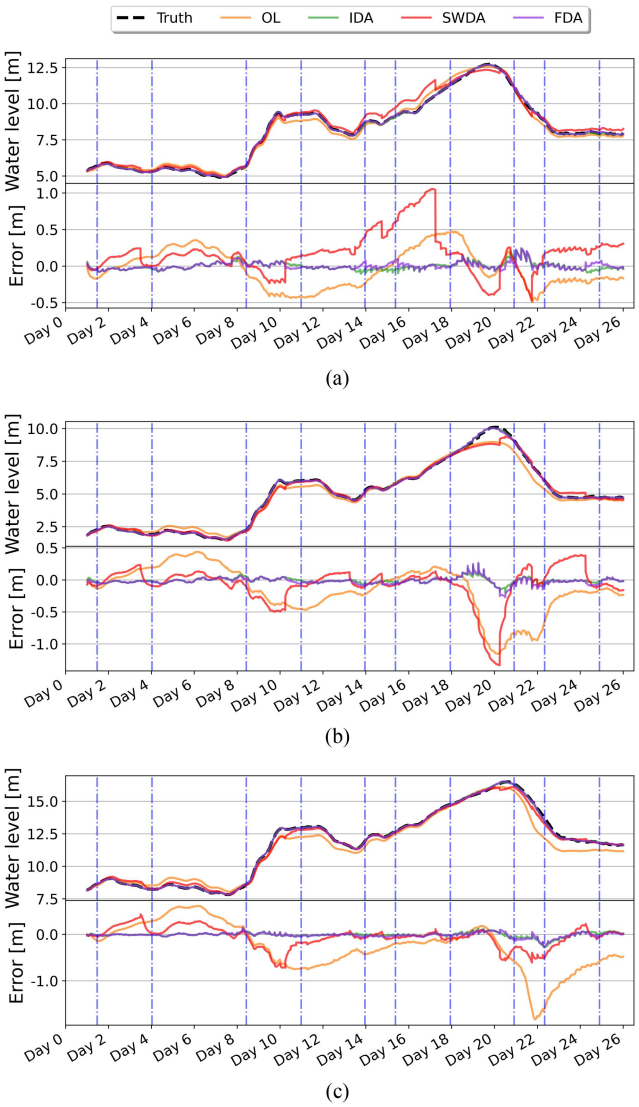


Fig. 8. WLs at observing stations (a) Tonneins, (b) Marmande and (c) La Réole for all experiments (top panels), and errors computed with respect to the *Truth* in-situ WLs (bottom panels).

At the TON station [see Fig. 8(a)], a discrepancy of 1 m was observed during the rising limb, highlighting once again the sensitivity of model accuracy to observation timing. This error was substantially reduced when the DA system assimilated the SWOT observation on day 18 (seventh vertical line in Fig. 8), reducing the WL errors to approximately 20 cm. The influence of observational timing is further illustrated at the MD0 station [see Fig. 8(b)], where the flood peak is underestimated by SWDA, as it occurred between two SWOT overpasses on day 18 and 21. Without timely observational updates, the model—constrained by previously estimated parameters—fails to capture the sharp rise in discharge.

Regarding the 2-D assessment of simulated flood extents compared to the *Truth* maps, Table III demonstrates that all DA experiments, especially IDA and FDA, exhibit improved performance and achieve higher CSI values relative to the OL simulation. Specifically, while the FDA provides superior results during the rising limb, the IDA attains the highest CSI at the flood peak on day 20. Although SWDA does not match the same level of WL accuracy at individual gauge stations, it nevertheless provides a strong representation of spatial flood dynamics, particularly during the rising and falling limbs. However, the absence of SWOT observations precisely at the flood peak limits the capability of SWDA during this critical phase, resulting in lower CSI values on day 20.

2) For shorter pass plans in OSSE: As previously discussed, the performance of the SWDA is strongly limited by the revisit frequency of SWOT, with current intervals proving insufficient to match the effectiveness of in-situ observations. Leveraging the OSSE framework, additional SWDA experiments were conducted to evaluate the potential benefits of future altimetry missions offering similar observational capabilities but with increased temporal resolution. Specifically, DA experiments were repeated under hypothetical revisit scenarios of 6, 12, 18, and 24 h, as described earlier.

It was shown that the SWDA experiments with increasingly denser SWOT observations are able to retrieve the *Truth* parameter values. Accordingly, as the revisit frequency decreases from 24 to 6 h, the SWDA experiments produce analyzed parameter values that progressively converge toward the *Truth*, approaching the accuracy achieved by IDA [green lines in Fig. 7(a)]. This is mostly true for all parameters, except K_{s_0} and K_{s_1}. While this could be expected for K_{s_0}, the analyzed values of K_{s_1} by different SWDA experiments reveal some problem with SWOT synthetical data. Indeed, regarding the floodplain friction K_{s_0}, the IDA and FDA in the previous experiments using the original pass plan also manifested the same pattern. In contrast, for K_{s_1}, the assimilation of SWOT observations—particularly those near the first meander of the TELEMAC-2D model domain after Tonneins, where the riverbed is governed by K_{s_1}—reveals signs of overbank flow into the adjacent floodplain. To maintain WLs consistent with these SWOT observations, both K_{s_0} and K_{s_1} are reduced, effectively delaying water evacuation from the floodplain and preserving inundation in this area over an extended period.

These results underscore the advantages of increased revisit frequency in future satellite missions. The setups with a 6 and 12-h revisit interval enable accurate retrievals of *Truth* parameter values and hydraulic states (except at the TON station), almost comparable to that achieved using high temporal sampling in-situ data (15-min intervals, as summarized in Table I) by IDA. Notably, across all four shorter pass plans, the impact on the discharge simulation is substantial, as evidenced by the close agreement between the reference discharge (black line) and the reconstructed time-series following the application of the corrected μ values in the SWDA experiments [bottom panel in Fig. 7(b)].

The key outcome of the OSSE results lies in the validation of assimilation robustness under known conditions, enabling a direct comparison against the *Truth* parameter values. The limitations of the SWDA experiment, due to the coarse temporal resolution of SWOT observations, were highlighted. This is reflected in the degraded estimation of friction parameters, notably during the flood peak [see Fig. 7(a)]. This has been significantly

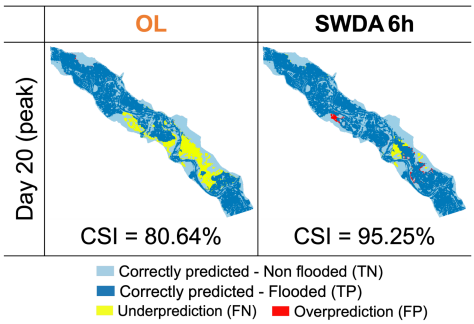


Fig. 9. Contingency maps for OL and SWDA 6 h experiments with respect to *truth* flood extents at flood peak (day 20).

reduced when the pass plan is enhanced with shorter revisit frequency. Comparing the SWDA under the original pass plan (one SWOT every 2–3 days) to the SWDA with a 18 h revisit frequency, the RMSE at the MD0 and LR0 observing stations decreases by 70.1% and 42.4% , respectively. It can also be asserted that revisit intervals shorter than 18 h begin to exhibit diminishing returns, offering no clear advantage relative to the increased volume of assimilated observations (in subdaily pass plans).

It is notable that the CSI values obtained from all four SWDA experiments with shorter pass plans are comparable (on day 17, 19, and 23) to or even exceed (on day 20) those achieved by IDA under the original pass plan. This highlights the substantial impact of SWOT revisit frequency on CSI performance, particularly evident during the flood peak on day 20, where the SWDA with a 6 h revisit interval attains the highest CSI (i.e., 95.25%). These findings underscore the value of SWOT observations in enhancing the 2-D representation of floodplain dynamics, while maintaining the accuracy of simulated WLs within the river channel (except at TON). The corresponding contingency maps of this experiment (i.e., SWDA with 6 h) are shown in Fig. 9 alongside that of OL, computed with respect to the *Truth* flood extent map at flood peak (day 20). The contingency maps display four possible outcomes for each pixel. Pixels where the model and observations match are shaded blue—light blue for nonflooded areas and dark blue for flooded ones. Pixels that were observed as flooded but predicted as nonflooded (underpredictions) appear in yellow, while pixels observed as dry but predicted as flooded (overpredictions) are shown in red. Fig. 9 highlights the improvement by SWDA with a 6 h revisit interval in representing the flood representation compared to OL, in the floodplain.

C. Results for Real Experiment

As previously outlined, the DA methods evaluated through OSSEs were subsequently applied to real SWOT observations acquired during the 2024 event. However, the prevailing meteorological conditions bring a major difference that is no overflowing in floodplain was observed in 2024. While the real SWOT data are not to be validated under flooded conditions, it is still relevant to assess their accuracy within the main channel. Unlike in OSSE, the *Truth* parameter values are not known beforehand

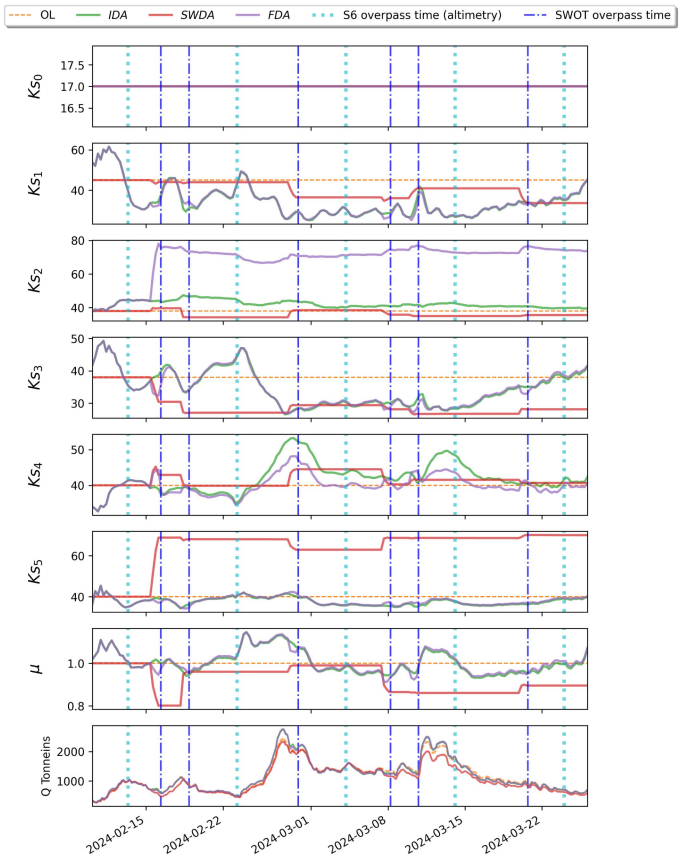


Fig. 10. Analyzed values of the control vector IDA (green), SWDA (red), and FDA (violet) in real 2024 event. Horizontal orange-dashed lines represent the default parameter values, while vertical dashed lines indicate SWOT and Sentinel-6 overpass times. From top to bottom: floodplain friction coefficient K_{s_0} (not controlled), riverbed friction coefficients K_{s_i} for $i \in [1,5]$, inflow correction factor μ, and upstream forcing $Q(t)$ and the simulated discharge time-series at Tonneins.

in the real event, precluding direct comparison against a reference. Therefore, the validations of the DA experiments rely on the use of in-situ, SWOT, and Sentinel-6 altimetry data.

1) Impacts on Control Vector: As in Fig. 7, the evolution of analyzed parameters from the different DA experiments for the 2024 real event is shown in Fig. 10. Horizontal orange dashed lines indicate the default or calibrated parameter values used in the OL, while green, red, and violet lines represent the IDA, SWDA, and FDA, respectively. Fig. 10 shows the temporal variation of the friction coefficients K_{s_i} for $i \in [1,5]$ (resulted from the zoning refinement of the riverbed). It should be noted that floodplain friction K_{s_0} is not included as a control variable in this real-data scenario. The bottom panel in Fig. 10 displays the upstream BC and the corresponding simulated discharge time series at the TON station, after applying the inflow correction factor μ.

The parameter estimates obtained from the IDA experiment remain within physically plausible bounds. In the case of SWDA, the analyzed parameters tend to remain closer to their initial values for a longer period due to the lower frequency of assimilation updates. Some notable deviations emerge when SWOT data is assimilated. For instance, the analyzed K_{s_2} from

TABLE IV
RMSE BETWEEN (A) SIMULATED WLS WITH RESPECT TO OBSERVED IN-SITU WLS OVER THE 2024 EVENT, AND SIMULATED WSES ALONG THE RIVER WITH RESPECT TO (B) SWOT AND (C) SENTINEL-6 WSE MEASUREMENTS

(a) RMSE [m] of (a) simulated WLs with respect to observed in-situ WLs. Italicized columns denote gauge stations only for validation in IDA and FDA.

Exp.	TON	LMA	MD0	*MD1*	COU	*LR1*	LR0
Stricker zone	K_{s_1}	K_{s_2}	K_{s_3}	K_{s_3}	K_{s_4}	K_{s_5}	K_{s_5}
OL	0.449	0.098	0.301	*0.213*	0.222	*0.224*	0.195
SWDA	0.491	0.201	0.261	*0.203*	0.346	*0.399*	0.383
IDA	**0.116**	**0.062**	**0.065**	*0.172*	**0.073**	*0.152*	**0.068**
FDA	0.127	0.079	0.073	***0.167***	0.100	***0.147***	0.080

(b) RMSE [m] of simulated WSEs with respect to SWOT-node WSEs.

Date time	2024-02-16 08:26:29	2024-02-18 21:45:26	2024-02-28 20:07:43	2024-03-08 05:11:32	2024-03-10 18:30:30	2024-03-20 16:52:45	Avg.
Pass	42	113	391	42	113	391	-
OL	0.910	0.631	0.506	0.713	0.663	0.568	0.665
SWDA	**0.562**	**0.371**	**0.260**	**0.372**	**0.238**	**0.295**	0.350
IDA	0.855	0.486	0.536	0.641	0.589	0.499	0.601
FDA	0.578	0.391	0.292	0.421	0.254	0.302	0.373

(c) RMSE [m] of simulated WSEs with respect to Sentinel-6 WSE measurements.

Date time	2024-02-13 08:07:14	2024-02-23 06:05:45	2024-03-04 04:04:16	2024-03-14 02:02:48	2024-03-24 00:01:21	Avg.
OL	0.289	0.321	0.340	0.347	0.330	0.325
SWDA	**0.270**	**0.307**	0.255	0.420	**0.293**	0.309
IDA	0.318	0.316	0.234	**0.206**	0.329	0.281
FDA	0.318	0.316	**0.233**	**0.206**	0.329	0.280

The lowest RMSE among the experiments for each column is boldfaced.

FDA, or K_{s_5} from SWDA, both can reach 75 $\mathrm{m}^{1/3}.\mathrm{s}^{-1}$. It should be reiterated here that the resolution of the inverse problem using the EnKF remains subject to equifinality, whereby different combinations of control parameters (here, K_{s_2} and K_{s_5}) may yield similar WL outcomes. As a result, the analyzed parameter values may exhibit compensatory behavior to reproduce the expected WLs, from the observing stations.

In this real event, with K_{s_0} excluded from the control vector due to the absence of floodplain overflow, the first riverbed friction coefficient K_{s_1} exhibits more stable behavior, showing reduced variability over the course of the event, particularly when compared to IDA and FDA. This contrasts with the OSSE scenario, where K_{s_0} and K_{s_1} frequently compensated for one another in response to equifinality effects. It is also worth-noting that while both IDA and FDA require a higher inflow discharge (with analyzed μ from IDA and FDA are mostly more than 1) SWDA consistently constrains the inflow throughout the event, beginning from the first assimilation of SWOT observations. The contrasting behavior observed between the upstream control variables (μ and K_{s_1}) and the downstream parameter (K_{s_5}) suggests that, through the assimilation of SWOT data, the SWDA tends to reduce the inflow volume while simultaneously favoring a more rapid outflow from the hydraulic domain.

2) Time evolution of WLs at observing stations: The performance of the DA strategies was evaluated at gauge locations. Table IV(a) summarizes the RMSE values computed between the simulated and observed WLs at each observing station over the full duration of the event. The italicized columns indicate stations used exclusively for validation, while the remaining stations were included in the assimilation for IDA and FDA. Fig. 11 compares the observed (black-dashed lines) and simulated WLs (colored lines) by the OL and DA analyses. Across all stations, both IDA and FDA experiments effectively capture the temporal evolution of WLs, with RMSEs generally below 17 cm. At all observation stations, the RMSE values obtained from FDA are, in general, slightly higher (i.e., 0.8 to 2.7 cm) than those from IDA. This suggests a potential conflict or inconsistency, which can be minor yet not negligible, between the in-situ and SWOT data sources assimilated in the FDA configuration.

The performance of SWDA, in terms of analyzed WLs at observing stations, is only comparable to that of the OL simulation. This stems from both the low revisit frequency of SWOT, which limits the regularity of observational updates needed for effective parameter correction in SWDA, and its incomplete spatial coverage of the domain by some SWOT passes (e.g., 42 and 113). This incompleteness shall be elaborated later in Section IV-C3. It can be noted that the SWDA analysis tends to underestimate WLs at most observation stations, notably at TON, LR1, and LR0. This underestimation primarily results from an overly low inflow discharge being introduced into the hydraulic domain, as previously illustrated in the bottom panel of Fig. 10.

Fig. 11 showed that both IDA and FDA experiments exhibit high accuracy in simulated WLs across the hydraulic domain—including at all seven gauge stations, with slightly higher RMSEs at stations used only for validation (e.g., MD1 and LR1). On the other hand, the SWDA experiment yields a noticeable improvement over the OL configuration only at the MD0 station. This localized enhancement results from the adjustment of the associated friction zone, where K_{s_3} by SWDA (red line) was reduced to approximately 28 $\mathrm{m}^{1/3}.\mathrm{s}^{-1}$ on Fig. 10.

3) Impact on WSE profile: The key strength of SWOT lies in its ability to provide spatially continuous WSE profiles along

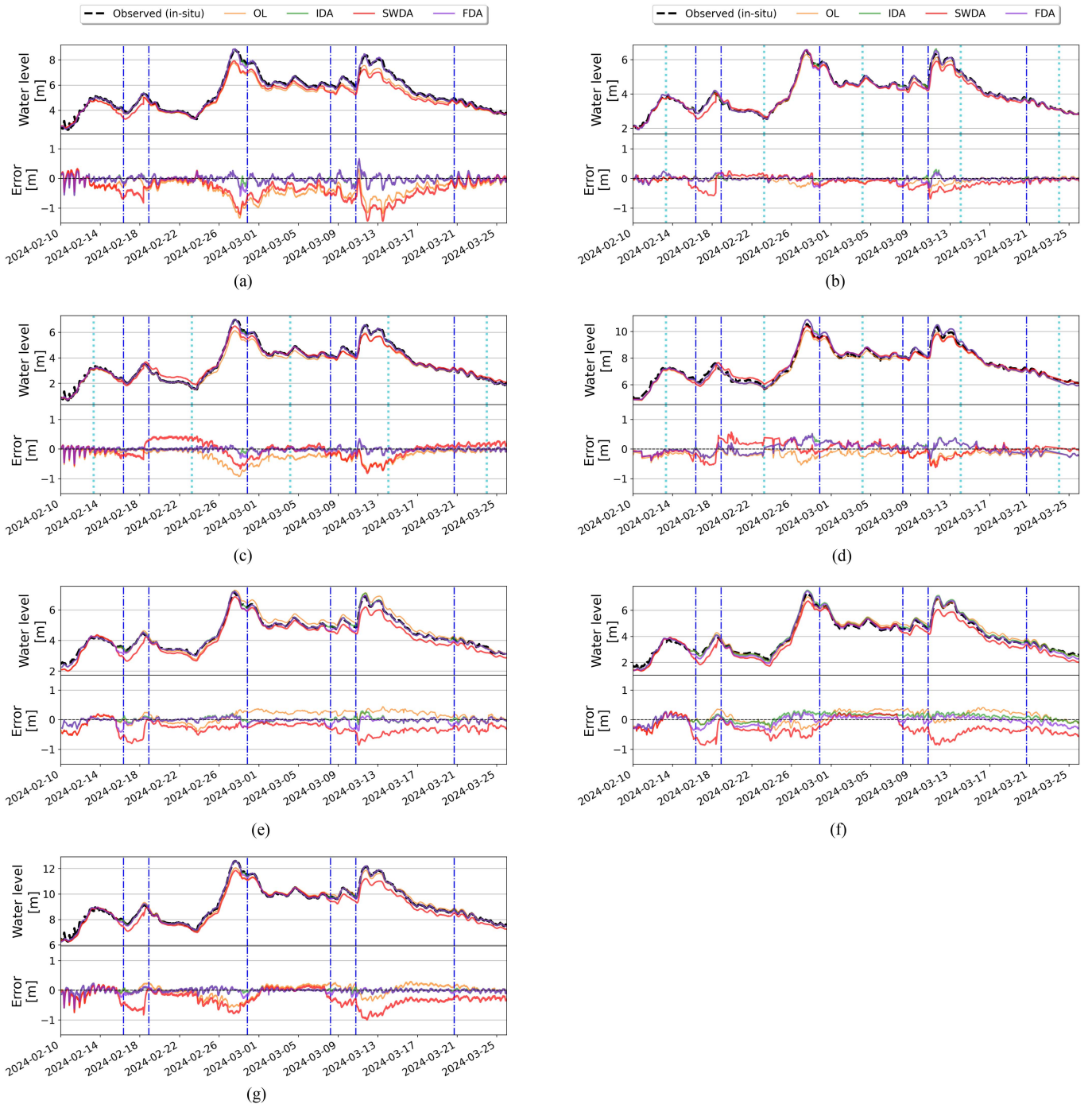


Fig. 11. WLs at observing stations for all experiments (top panels), and errors computed with respect to the observed in-situ WLs (bottom panels). (a) Tonneins (TON). (b) Le Mas-d'Agenais (LMA). (c) Marmande (MD0). (d) *Marmande 1 (MD1)*. (e) Couthures-sur-Garonne (COU). (f) *La Réole 1 (LR1)*. (g) La Réole (LR0).

the river for each satellite pass. Fig. 12(a), 12(b), and 12(d) present the SWOT node observations alongside equivalent WSEs simulated by the DA experiments. It is important to note that, unlike in the OSSE framework, portions of the river between TON and MD0, as well as between TON and LMA, are not covered in Fig. 12(a) and (b), respectively, due to these sections falling outside the swath coverage of passes 42 and 113.

Fig. 12(a) shows that the WSEs along the river's downstream part are overestimated by the OL (orange line) and IDA (green line), in the river segments governed by K_{s_4} and K_{s_5}. In contrast, SWDA (red line) and FDA (violet line) maintain close to the SWOT observations. Yet, to achieve this consistency, SWDA compensated by significantly reducing the inflow discharge (adjusting μ to 0.8), while also increasing K_{s_5} to 75 $\mathrm{m}^{1/3}.\mathrm{s}^{-1}$. This behavior indicates a tradeoff in SWDA, where performance

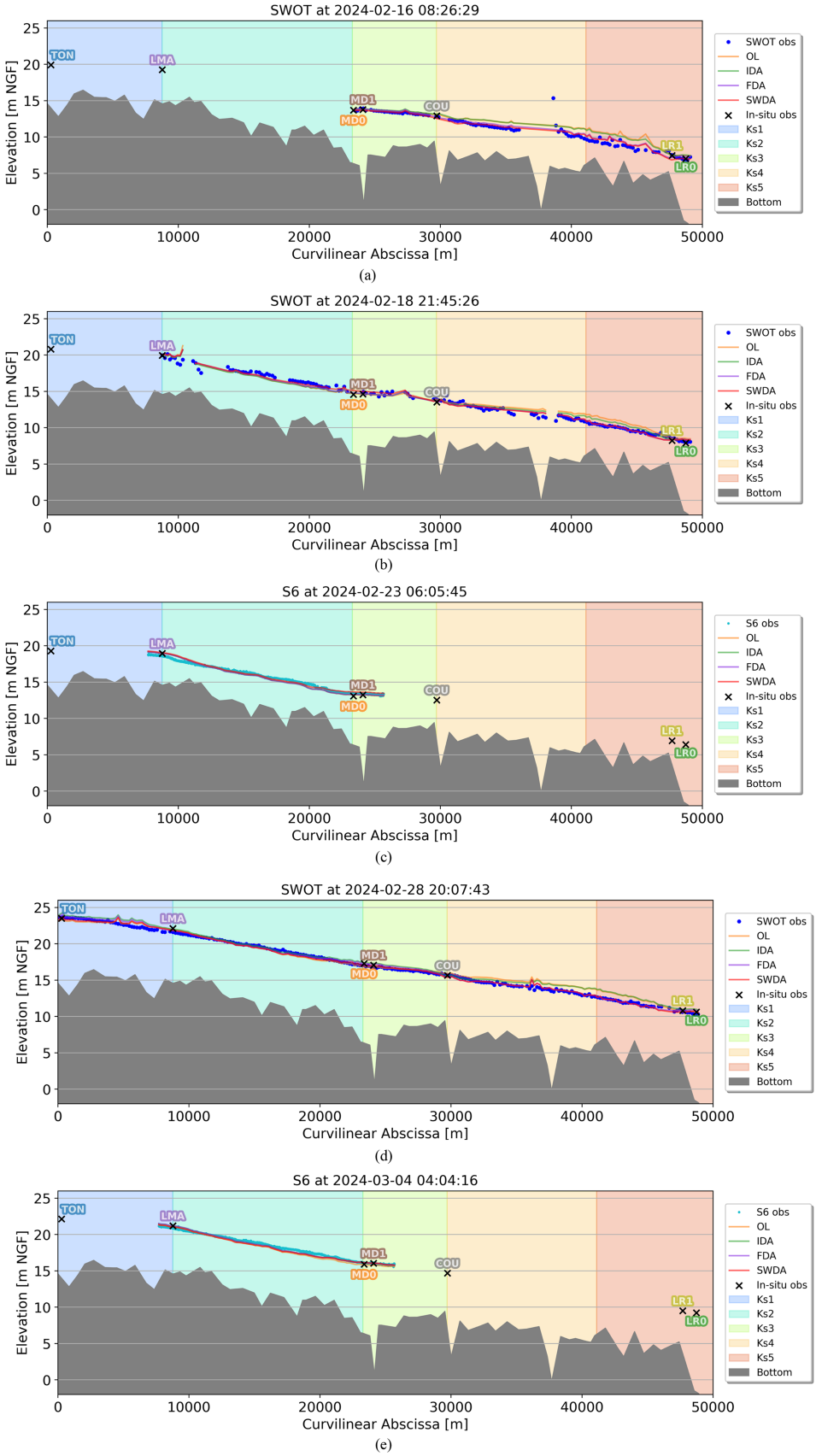


Fig. 12. WSE profiles between observed SWOT nodes (blue dots) or S6 FFSAR measurements (cyan dots) and OL (orange), IDA (green), SWDA (red) and FDA (violet). Circles indicate in-situ WSEs measured at gauge stations along the river reach. (a) SWOT observation on 2024-02-16 (pass 42). (b) SWOT observation on 2024-02-18 (pass 113). (c) Sentinel-6 observation on 2024-02-23. (d) SWOT observation on 2024-02-28 (pass 391). (e) Sentinel-6 observation on 2024-04-03.

at gauge stations (Table IV(a)) is sacrificed in favor of improved consistency with SWOT observations along the river profile. Another similar situation occurs during pass 42 on 2024-03-08, where SWDA reduces the inflow correction factor μ to 0.9 in order to better align with the partially observed WSE profile from SWOT nodes. On the other hand, by leveraging both in-situ and SWOT data, the FDA experiment maintains strong performance across the metrics in the observation space, without the need to compromise between accuracy at gauge stations and alignment with SWOT-derived profiles.

The RMSEs computed against SWOT WSEs at river nodes are summarized by Table IV(b). It was shown that while IDA reduces errors at the gauge locations, it still produces overestimations of the WSE profile across large portions of the domain. In the downstream part, this discrepancy is likely related to uncertainties in the representation of the main channel bathymetry. As previously illustrated in Fig. 2, the actual up-to-date riverbed appears to lie below the elevation defined in the TELEMAC-2D model. In addition, the lack of gauge station at the transition between K_{s_4} and K_{s_5} zones can also lead to overestimated WSEs, as both La Réole stations (LR0 and LR1) are positioned near the downstream boundary of the K_{s_5} zone. In contrast, the assimilation of all SWOT river nodes enables a more accurate estimation of the friction coefficients required to reproduce the observed WSE profiles. While bathymetric imperfections persist, they are partially compensated by elevated Strickler values, in some case exceeding 70 $\mathrm{m}^{1/3}.\mathrm{s}^{-1}$. This adjustment reflects the well-known equifinality between friction and bathymetry in hydraulic modeling, as previously discussed in [63].

The added value of SWOT observations becomes evident when compared to in-situ data, as they enable correction of the entire longitudinal water surface profile. In the SWDA experiment (red lines), a minor discrepancy persists at the downstream end, as only K_{s_5} and μ can be modified to be consistent with the downstream rating curve. Water surface profiles from the FDA experiment (violet lines) are shown closely mirror those from SWDA, as also reflected by the RMSEs in Table IV(b). However, such strong performances depend on the availability and distribution of SWOT observations along the river. While the SWOT revisit frequency remains a limitation, the results presented in Figs. 11 and 12 (FDA in violet) clearly demonstrate that combining in-situ measurements with SWOT-derived observations significantly enhances the accuracy of water surface profiles. This consistency is also expected to be maintained under both low- and high-flow conditions, as demonstrated in the OSSE.

To support the assessment of SWOT-derived WSEs and the performance of the DA framework, independent validations were also conducted using Sentinel-6 altimetry data (processed with FFSAR) which observe the middle part of the domain between LMA and MD1. As such, Fig. 12(c) and 12(d) present the comparison between Sentinel-6 WSE profiles (in cyan) along the river centerline, on 2024-02-23 and 2024-03-04, and the WSE profiles from the DA analyses. The RMSEs computed against all Sentinel-6 WSE profiles along the river centerline for all experiments are summarized by Table IV(c). Overall, a moderate agreement is observed between the Sentinel-6 WSE measurements and the simulated WSEs along the river centerline from the OL, with discrepancies reaching up to 34.7 cm. Similar discrepancies are noted in the IDA for several dates, although notably lower RMSE values, ranging between 20 and 23 cm, are achieved during higher flow conditions on 2024-03-04 and 2024-03-14. Some pronounced deviations in the SWDA results are evident on 2024-02-23 and 2024-03-14, following the assimilation of SWOT observations from pass 113 on 2024-02-18 and 2024-03-10, respectively. These discrepancies likely arise from the missing SWOT coverage between the TON and LMA stations (also sparse around LMA), as illustrated in Fig. 12(b), resulting in assimilation predominantly guided by SWOT nodes near the downstream station. Moreover, since only a single friction coefficient (K_{s_2}) governs this river segment with respect to the Sentinel-6 WSE profile, the lack of comprehensive SWOT data along the full reach restricts SWDA's capacity to effectively reduce errors across the entire stretch between LMA and MD1.

In summary, the comparison between IDA and SWDA, or in other words between in-situ and RS observations, highlights the complementary strengths of each data source. In-situ measurements offer reliable WL data with high temporal sampling at discrete locations, yet they are limited in their ability to capture the continuous water surface profile between stations. While this limitation could be addressed by increasing the density of gauge stations, the associated costs are often prohibitive. Conversely, RS-derived WSE data (either from SWOT or Sentinel-6) provides large spatial coverage observations upon a catchment, albeit at lower temporal resolution. Results from the OSSE indicate that a denser revisit interval (ideally up to daily or subdaily) would be sufficient to support high-quality reanalyses; however, such a satellite constellation is not currently operational. This combined strategy, as shown by FDA, enables robust temporal and spatial interpolation, effectively bridging the limitations of individual data sources. The findings presented here advocate that the proposed methodology—integrating model and multisource data—can achieve the accuracy required for operational applications such as water balance estimation and flood risk management.

V. Conclusion

This study highlights the merits of integrating satellite altimetry data—particularly from the recently launched SWOT mission—as a transformative resource for improving the representation of riverine flood hydrodynamics. Here, we employed multisource DA within the framework of an EnKF integrated with the TELEMAC-2D hydrodynamic model over a reach of the Garonne River. Through various DA experiments conducted using both synthetic and real observational data, we have highlighted key findings that could inform future flood reanalysis and forecasting systems.

Results from the OSSE confirm the ability of EnKF methods to improve model parameter estimation, especially when assimilating high-temporal-resolution in-situ gauge data, as demonstrated in the IDA experiment. In contrast, strategies relying solely on SWOT data (SWDA) were limited by the temporal resolution of satellite overpasses, leading to challenges in constraining hydraulic parameters during flood events. The assimilation of both in-situ and SWOT-derived WSE data (FDA) proved to be

the most effective strategy, offering robust parameter retrieval across a range of hydraulic conditions.

Given the absence of an overflow event over the Garonne River since its launch, the 2024 event was examined with a focus on SWOT's capabilities in monitoring the riverbed. This assessment extended beyond pointwise RMSE at gauge stations to include the longitudinal WSE profile, providing a more comprehensive evaluation of model performance with and without SWOT data. The nature of the 2024 event explains the apparent lack of clear assimilation efficiency over the floodplain (compared to those using Sentinel-1 flood extent maps during major flood events, as presented in our previous studies [23], [24]). It is not necessarily indicative of a systematic issue with SWOT data itself. On the other hand, although the event did not include overbank flows, assimilation of SWOT data in combination with in-situ WL data successfully reproduced river dynamics, showing promising prospects for future flood monitoring systems. Independent validation using Sentinel-6 measurements (only used for validation) further offers a more comprehensive assessment of model performance in simulating WSEs along the river, even when only SWOT-derived data were assimilated (in SWDA). These findings emphasize SWOT's added value relative to the OL simulation and highlight the growing potential of satellite altimetry for real-time flood risk management, particularly in ungauged basins.

Nevertheless, it should be emphasized that the system's efficiency remains sensitive to the DA settings in accordance with the catchment characteristics and flow dynamics. In particular, the definition of the control vector should reflect the main sources of uncertainty in the 2-D hydraulic model, as well as those associated with its input data. In addition, the DA window length should be tailored to the specific flow dynamics of the study area, since different hydrological regimes may require different assimilation time scales. These aspects are crucial for ensuring the robustness and reliability of the methodology across diverse hydrologic catchments.

This study lays the groundwork for future investigations of overflowing events using SWOT PIXC products. In forthcoming work, we plan to incorporate PIXC data alongside other RS-derived flood extent products, e.g., from Sentinel-1 and Sentinel-2. This is expected to enable a more robust evaluation of SWOT's potential for floodplain representation. As such, the study paves the way for future research exploring the integration of additional data sources, such as high-resolution flood extent observations or real-time sensor networks, to further refine flood forecasting models. As satellite altimetry technology continues to evolve, e.g., new hydrology-dedicated missions like SMASH [64], the prospects for integrating such data into operational flood management systems become even more promising, offering a pathway to more accurate and timely flood predictions.

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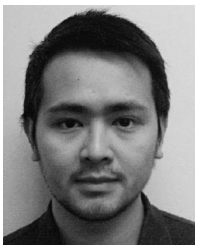
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Subsequently, he has worked on environmental acoustics with the support of a grant from the Fondazione Confalonieri. He has been appointed as a Professor of numerical analysis with the University of Milan, Milan, Italy, in 1994. In 1997, he moved to CERFACS, Toulouse, France, where he has led the development of a dynamic coupler oriented toward the modular implementation of data assimilation (DA) operational suites. He has been a Visiting Scientist with EPCC, Edinburgh, U.K., the University of Corvallis Oregon, Corvallis, OR, USA, the ECMWF, Reading, U.K., and the CMCC, Bologna, Italy. Since 2007, he has been a Scientific Consultant in different fields of application of operational DA.



Sophie Ricci received the engineering degree in computational fluid dynamics and environmental sciences from the École Nationale Supérieure d'Électrotechnique, d'Électronique, d'Informatique, d'Hydraulique et des Télécommunications, Toulouse, France, in 2000, the M.Sc. degree in environmental sciences from the Institut Polytechnique de Toulouse, Toulouse, in 2000, and the Ph.D. degree in ocean physics from Université Paul Sabatier, Toulouse, in 2004.

During her Ph.D. work with CERFACS, Toulouse, she investigated a multivariate formulation of the background error covariance matrix for variational assimilation in global ocean. Since 2008, she has been with the Climate, Environment, Coupling and Uncertainties Team, CERFACS/CNRS UMR 5318, Toulouse. Her research interests are focused on uncertainty quantification and reduction, with data assimilation algorithms in geosciences, with a strong focus on river hydrodynamics and flooding.

Charlotte Emery received the Graduate degree in physical numerical modeling from the National Institute of Applied Sciences (INSA), Toulouse, France, in 2013. She received the Ph.D. degree in continental interfaces and hydrology from the University of Paul Sabatier, Toulouse, France, in 2017.

Her thesis focused on studying the contribution of the future wide-swath altimetry mission Surface Water and Ocean Topography (SWOT) for large-scale hydrological modeling. Following her Ph.D., she undertook a postdoctoral fellowship at the Jet Propulsion Laboratory in Pasadena, continuing her work in the same thematic areas. After a six-year career in industry as an engineer in satellite Earth observation data processing, she joined the National Centre for Space Studies (CNES) as a permanent researcher in 2025. Her current missions revolve around the hybridisation, distribution, and valorisation of spatial hydrology data.



Christophe Fatras received the Ph.D. degree in space altimetry with CNES/ONERA/GET, Toulouse, France, in 2015.

He worked for two years in hydrology and glaciology from space remote sensing data at ESPOL (in Ecuador), in particular for flood monitoring purposes. He also worked at CESBIO laboratory on the hydrology of the Congo Basin from SMOS data. He develops since 2019 innovative algorithms for flood monitoring, water surface detection and water volume estimation from various remote sensing data at CLS.



Santiago Peña Luque is an Engineer in telecommunications specialized in scientific space payloads (European "Space-Master" program), 2009, Luleå University of Technology. Since joining CNES in 2010, he has contributed to the developments of generic tools for telemetry mission and monitoring centers (MSL Curiosity, Rosetta-Philae, Insight, etc.). In 2017, he joined the SWOT-Downstream-team to improve the integration of satellite measurements in the world of hydrology (modeling, applications). Such activities are focused on water surface detection using satellite images and land cover classification to improve water stock and river run-off estimations.



Raquel Rodriguez Suquet received the engineering degree in telecommunications from the University of Catalonia, Barcelona, Spain, in 2004.

Since 2005, she has been with the Centre National d'Etudes Spatiales (CNES) as a permanent staff. She is a Specialist Radar Engineer with more than 15 years of experience. She was in charge of the radar performance and instrument design of active and passive sensors of several space missions. In 2022, she joined the Earth Observation Laboratory Department, CNES, where she is In Charge of developing new earth observation applications and solutions to support users using satellite data.